

Expected-utility representation $\Rightarrow$ Expected multi-utility representation

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Expected-utility representation.** There is a function $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$.

0.2 Conclusion

- **Expected multi-utility representation.** There is a set U of functions $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$ for every $u \in U$.

0.3 Proof

Take $U = \{u\}$.

Record: topics/decision-theory/results/eu-implies-multi-utility.yaml